

Photorealistic Synthetic Human Face Generation Using Adversarial Network cDCGAN and StyleGAN With Text-Guided Feature Control

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Abstract - Over the last several years, the area of image generation has evolved into deep learning, which allows producing extremely realistic and various visual content with a model. Nevertheless, systematic comparison of architectures is essential despite all these developments to evaluate how effective they are. This paper contains a comparative analysis of StyleGAN and Conditional Deep Convolutional GAN (cDCGAN) which are popular generative images network models. Both of them were trained on image datasets with PyTorch and with a combination of preprocessing and augmentation techniques to enhance the stability of the training and the diversity of the output. The models have been tested by a qualitative inspection and quantitative analysis by means of Frechet Inception Distance (FID) and Inception Score (IS). The experimental findings show that StyleGAN is better than cDCGAN with lower FID of 30.72 and higher IS of 17.83 which indicates better image quality and more realistic alignment to the real data distributions. On the contrary, cDCGAN had mode collapse and lower diversity, which shows its inability to capture finer data variations. In addition to raw metrics, StyleGAN also had a smoother convergence curve during training and produced images that had more details and more coherent structures. This paper highlights the significance of model selection in generative models and offers empirical data that StyleGAN is more appropriate to produce high-quality images than cDCGAN. The results aid the development of a greater insight into the trade-offs between standard GANs and advanced architectures, which can be used fruitfully in future studies in the field of computer vision, creative AI applications, and domain-specific generative tasks.

Keywords – Computer vision, Conditional DCGAN, StyleGAN, Text Encoder, CLIP

I. INTRODUCTION

Realistic face generation is now a significant application of deep generative models to computer vision [5], [6]. Generative Adversarial Networks (GANs) have become one of these frameworks that have made significant contributions to the behavior of high-quality image generation [5], [6]. A GAN is made up of two networks, one is the generator, which generates synthetic images, and the other one is the discriminator, which determines whether an image is a real

image or a generated image [5]. With this adversarial training, the generator is steadily enhanced, which becomes more realistic over time [5], [6].

Conditional GANs are an extension of this model, where auxiliary information, like class labels or text queries, are used to direct the generation process [3], [4], [8]. One successful variant, the conditional Deep Convolutional GAN (cDCGAN), uses random noise generated by a Gaussian distribution in addition to text embeddings created by user prompts [3], [8]. The generator gradually upsamples a low-resolution image (e.g., 4×4 pixels) to greater resolutions (e.g., 32×32 pixels), and the discriminator downsamples both the generated and real images to evaluate authenticity [5], [3]. During training, it is divided into epochs and batches as the model is repeatedly trained to optimize its parameters in order to more closely approximate the generated images with the conditioning information [3], [8].

StyleGAN has more recently established a new standard in face synthesis with images of high resolution and photorealism [4], [5], [1]. As compared to the previous GANs, StyleGAN presents an intermediate latent space (W), which enhances the disentanglement of features [4], [5]. Using Adaptive Instance Normalization (AdaIN), the style based on the latent vector is added to every layer of the synthesis network, and a particular feature, like facial shape, skin texture, and expression, can be fine-tuned [4], [5]. It also causes stochastic noise at various scales to add natural variations, e.g., wrinkles or freckles [5]. StyleGAN, instead of explicitly manipulating the input latent vector, starts with an effective learned constant ($4 \times 4 \times 512$) and upsamples the image through convolutional layers until the image has reached a resolution of 1024×1024 pixels [5]. These outputs are evaluated by the discriminator by successively downsampling the output, so that the generator is continuously enhancing image quality [5].

This paper is about creating synthetic human faces with the help of both cDCGAN and StyleGAN where user prompts are the conditional inputs [3], [4], [8]. In comparing the two architectures, we want to analyze how effective they are in generating realistic and semantically accurate facial images

[1], [4], [5], [7]. The discussion points out the strengths and weaknesses of each of these methods, giving an understanding of how conditional embeddings, latent representations, and adversarial training contribute to the overall improvement in quality and controllability of face synthesis [3], [4], [5], [8].

II. RELATED WORKS

Conditional GANs (cGANs) have also been applied to synthesize facial images based on text descriptions and allow images to be generated contingent upon auxiliary information (text embeddings) [8], [9]. The original research used a conditional Deep Convolutional GAN (cDCGAN) as a text-to-face synthesis model with the input being a text description of a face and the output being the generated image based on that input [8], [9].

Since GANs need numerical representations of text, the paper used two text encoders: Sent2Vec and RoBERTa [9]. These encoders were trained on a self-generated Spanish corpus based on CelebA dataset captions, which is the preprocessing phase referred to as corpus creation [9]. In the case of Sent2Vec, sentences were sanitized and combined into a large training corpus [9]. In the case of RoBERTa, sentence pairs were run through a Siamese network and a similarity score (0–1) was obtained on English text with SpaCy and compared to the Spanish versions [9]. About 250,000 sentence pairs were found in the RoBERTa+CelebA corpus [9]. Sent2Vec encoder was trained on the Spanish corpus during 7 hours, and RoBERTa encoder was fine-tuned during 42 days on the Spanish corpus with a loss based on the cosine similarity. Such trained encoders gave sentence embeddings that numerically encoded the descriptive text and could serve as conditioning vectors to the generator.

During the image generation stage, the cDCGAN took such embeddings as the conditions to generate the images based on the textual descriptions [8], [9]. The discriminator was trained to discriminate between the actual CelebA images and the fake images, which enhances the realism of the generated images [8], [9]. The quality of the generated images was measured using evaluation metrics like Inception Score (IS), Frechet Inception Distance (FID), and LPIPS (perceptual similarity) [9]. It was demonstrated that the RoBERTa+CelebA model performed better than Sent2Vec+CelebA and the control RoBERTa, generating sharper images with higher semantic consistency and more visual realism [9].

Face generation generative adversarial networks (GANs) and their derivatives have also been widely tested on the task of face synthesis [5], [6]. The first attempts at using standard GANs defined the adversarial training process, yet the generated images were of limited realism and frequently unstable [5], [6]. This concept was later extended to conditional GANs (cGANs) that condition the generation process with auxiliary data (class labels or text prompts) to produce more controlled outputs [3], [8], [9]. The quality was further enhanced by Deep Convolutional GANs (DCGANs) due to their use of convolutional layers, although artifacts and low-resolution results were still a problem [5], [6].

Subsequent improvements made StyleGAN redefine face synthesis, where the latent space is mapped to an intermediate space (W space) and the style injected at various synthesis layers through Adaptive Instance Normalization (AdaIN) [4], [5]. This structure enabled resolution and realism to be greatly enhanced and enabled the fine-grained control of attributes

such as facial structure, skin texture, and expression [4], [5]. Research continues to demonstrate that StyleGAN has lower Frechet Inception Distance (FID) and higher Inception Scores (IS) than previous GAN architectures, confirming its ability to create photorealistic and diverse faces [1], [4], [5].

More recent work has also added text encoders (e.g., CLIP, Sent2Vec, RoBERTa) to enhance text-to-image alignment in conditional systems [8], [9]. Comparative works find that robust text encoders enhance semantic fidelity and controllability, and advanced GAN architectures such as StyleGAN are the current state of the art in generating high-quality synthetically generated faces [4], [5], [9].

III. DATASET DESCRIPTION

SFHQ-T21 data set is a set of synthetically created images of human faces that should facilitate text-to-image generation and evaluation research. The dataset has a realistic face image in each entry with the associated descriptive text prompt and generation metadata. The dataset will serve to offer a systematic standard in investigating prompt-image matching, generative quality, and controllability in contemporary diffusion and transformer-based models.

Each image thus generated is stored under a distinct file name identifier to make it easily indexed and reproducible. All of the images in this dataset were synthesized by DALLÉ-3, a high-end text-to-image generative state-of-the-art model that has demonstrated itself to be capable of synthesizing photorealistic human faces [1], [5]. Each image includes the original prompt in English that was used to generate it. With every prompt-image pair, we store metadata (in structured form, in JSON) documenting the image generation parameters. This includes:

- **Resolution:** The size of the generated image (e.g. 512x512, 1024x1024).
- **Quality:** The level (or fidelity level) of refinement given when generating.
- **Style:** Characterizes any stylistic features used (e.g. photorealistic, studio portrait, natural lighting).

A. Generative Models Used in SFHQ-T21

DALLÉ-3: A generative model created by OpenAI that is based on the use of a transformer and is commonly known to generate face images that are both highly photorealistic and semantically accurate.

FLUX1_dev: This is a developmental version of the FLUX family, designed to be used to experiment and fine-tune text-to-image pipelines at their initial stages.

FLUX1_pro: An enhanced version of the FLUX model, which is trained and produces more realistic and more consistent face images than the development version.

FLUX1_schnell: FLUX1 is a fast, lightweight version of FLUX model optimized to be efficient in real time or in large scale tasks of generating content at comparatively lower computational cost.

SDXL (Stable Diffusion XL): It is a powerful diffusion model that is characterized; it is controllable and produces high-resolution images and it has been widely adopted by the research community. A summary of image production by both the models is presented in Fig. 1.

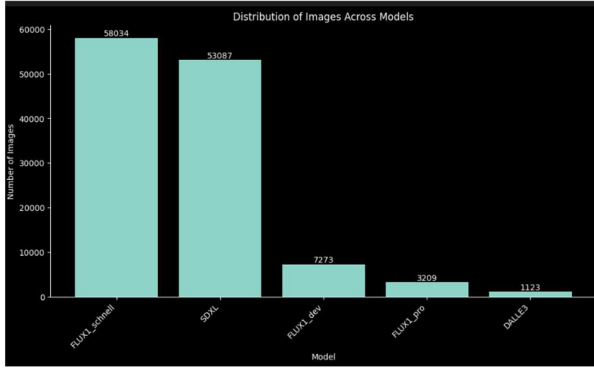


Fig. 1. Analysis of image generation by each model

B. Prompt length statistics in SFHQ – T21

Prompt length has a huge effect on the quality of generated images. Short prompts (approximately 15-25 words) tend to only extract higher-level features like age or gender leading to less detailed results. More detailed descriptions are given by longer prompts (more than 60 words) although they can create redundancy and inconsistencies. The mean prompt length in the SFHQ-T21 dataset was 49.5 words with the majority of prompts being between 40-57 words. The summary of this distribution is presented in Table 1. This was the middle range that gave us the best balance, more semantically accurate and face image consistent than either very short or too long prompts. The difference in prompt length is shown in Fig. 2.

Metric	Word Count	Character Count
Minimum	16	84
Maximum	164	1018
Mean	49.5	310
Median	47	299
25 th Percentile	40	248
75 th Percentile	57	363

Table 1: Word and Character Count

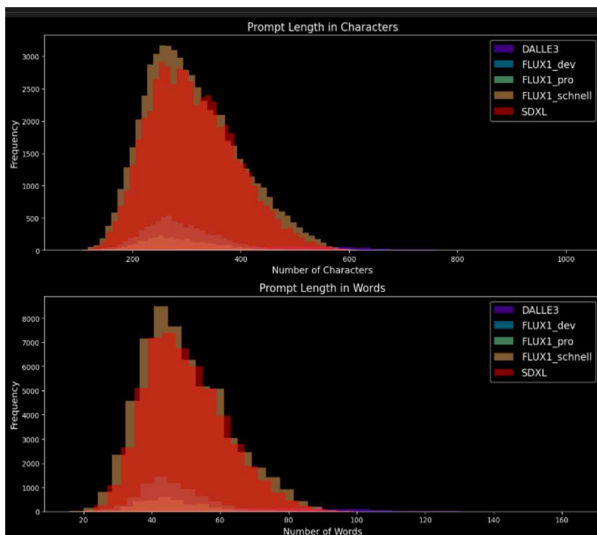


Fig. 2. Graph representing the prompt length

IV. PROPOSED METHODOLOGY

A. Preprocessing and Exploration of Dataset

The data is made up of artificially created human face images, associated descriptive text prompts, and structured metadata [1], [5]. Every entry was named by file name and tagged by the model used to generate it (DALLE-3, SDXL, FLUX1-dev, FLUX1-schnell). Preprocessing of text prompts consisted of eliminating inconsistencies and preparing them for use with statistical analysis [8], [9].

B. Prompt Analysis

Short lengthening was calculated based on the number of words and the number of characters [8], [9]. The descriptive statistics (minimum, maximum, mean, median, interquartile ranges) were computed [8], [9]. To depict the differences in complexity of prompts between models, visualizations such as histograms and distributions were plotted [8], [9]. This analysis provides an idea of the extent to which the textual guidance is detailed and descriptive for generating images [8], [9].

C. Visualization and Categorization Model-Based

The dataset was subdivided based on the model used to generate each image [8], [9]. Distribution plots were created to demonstrate the images generated by each model [8], [9]. Each model included representative samples, their respective prompts and metadata, and offered qualitative information on prompt-image alignment [8], [9].

D. Text Encoder

In text-to-image generative models, it is important to process textual descriptions into meaningful representations to direct the image generation process [8], [9]. One such effective tool is the CLIP (Contrastive Language-Image Pretraining) text encoder, since it can generate semantically rich embeddings that are consistent with visual concepts [9].

This is done by tokenizing the input text and sending it through the transformer-based text encoder of CLIP [9]. The encoder produces a fixed-size vector embedding containing the semantic content of the text prompt. The embedding is then trained with the generator in the GAN, so that the model can generate images that match the text input [8], [9].

Various integration approaches are typically used: with conditional GANs, the text encoding is appended to the latent noise vector to direct image generation [8], [9]. In attention-based architectures, like AttnGAN, the embedding applies attention maps to the generator, which uses them to target important text features at various synthesis stages [8], [9]. Moreover, CLIP embeddings can also be used to define loss functions that promote correspondence between generated images and input text in the shared embedding space [9].

The application of CLIP embeddings has the following benefits: it can capture finer semantic details, enable cross-modal matching of text and images, though it requires large-scale pretraining on diverse image-text pairs [9]. Overall, adding CLIP text encoders to GAN-based models greatly improves the quality and semantic fidelity of generated images [8], [9], [5].

E. Architecture of conditional DCGAN and StyleGAN

- Conditional DCGAN Architecture

The architecture of Conditional DCGAN training has a starting step of sampling z latent variable of a normal distribution together with a class labeling [8]. The generator receives the inputs and generates a synthetic image given the label [8]. Simultaneously, an actual image of the same label is taken out of the dataset. Both real and generated images are provided to the discriminator along with labels and a loss is calculated based on how well it classifies real and fake, given the label [8], [5]. This loss is minimized to update the discriminator. The next latent vector and label are then sampled and the generator generates another conditional image to be real is calculated and sent to the generator, which is updated to reduce this loss [8]. This iterative training process proceeds until convergence, and the generator is able to generate class-conditioned images that are very similar to real data distributions [8].

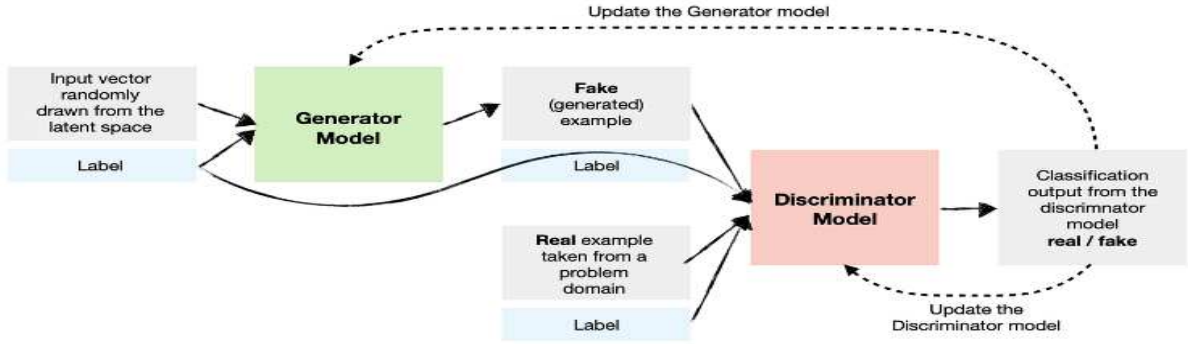


Fig. 3. Conditional DCGAN Architecture

- StyleGAN Architecture

StyleGAN training Andrew et al. (2019) [6] start with a sampling of a latent vector z , distributed as normal distribution. This is fed through a mapping network that converts it to an intermediate latent space representation w [6]. A synthesis of an image of w is then done by the generator [6]. At the same time real images are sampled in the set. The discriminator is compared to both real and generated images and its loss is calculated as the difference between the log-probability of real images and the log-probability of generated images [5], [6]. The discriminator is revised so as to reduce this loss. The second step involves sampling and mapping a new latent vector to w , generating another generated image [6]. The loss of the generator is the log-probability of the discriminator classifying this image to be real, and the generator and mapping network are updated based on this loss [5], [6]. This iterative process is repeated until convergence, which leads to a generator that is able to generate high-quality and realistic images [6].

In StyleGAN, a latent vector is first transformed into an intermediate latent representation w through a mapping network F as shown in Eq. (4):

$$w = F(z) \quad (4)$$

Conditional DCGAN is a variant of vanilla GAN that conditions both the generator and the discriminator on the labels of the classes. The generator generates counterfeit samples as a result of a latent variable z and a class label y , as shown in Eq. (1):

$$x_{fake} = G(z, y) \quad (1)$$

The discriminator receives both images and their associated labels as inputs. Its loss is defined as shown in Eq. (2):

$$L_D = -E_{\{(x_{real}, y) \sim p_{data}\}} [\log D(x_{real}, y)] - E_{\{z \sim p(z), y \sim p(y)\}} [\log(1 - D(G(z, y)))] \quad (2)$$

Similarly, the generator is trained to maximize the discriminator's probability of classifying generated images as real, with the loss function, as shown in Eq. (3):

$$L_G = -E_{\{z \sim p(z), y \sim p(y)\}} [\log D(G(z, y))] \quad (3)$$

The generator synthesizes an image from this intermediate latent code as shown in Eq. (5):

$$x_{fake} = G(w) \quad (5)$$

To train this discriminator, the objective is to maximize its ability to distinguish real images from generated ones. The discriminator loss is defined as shown in Eq. (6):

$$L_D = -E_{\{x_{real} \sim p_{data}\}} [\log D(x_{real})] - E_{\{z \sim p(z)\}} [\log(1 - D(G(F(z))))] \quad (6)$$

The generator, on the other hand, aims to fool the discriminator. Its loss is given as shown in Eq. (7):

$$L_G = -E_{\{z \sim p(z)\}} [\log D(G(F(z)))] \quad (7)$$

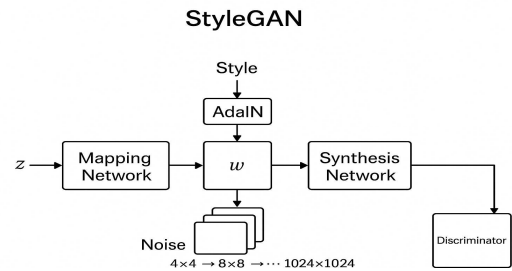


Fig. 4. StyleGAN Architecture

In addition to face synthesis, StyleGAN has been effectively modified for texture generation in industrial applications and medical image enhancement. Fig.4 represents the StyleGAN architecture. These examples emphasize the flexibility of StyleGAN to the application in areas where both the structure and the detail of the images are essential. Moreover, it can generalize and, as a result, can become a powerful candidate in multimodal and domain-specific generative tasks in the future.

V. IMAGE GENERATED BY MODELS



Fig. 5. Image generated by Conditional DCGAN

Fig. 5. illustrates the sample face image which are created by Conditional DCGAN model. Individual images are conditioned with a text prompt that defines factors like age, gender, hairstyle or facial expression. The results show that Conditional DCGAN can learn simple semantic properties of text prompts, but finer-grained information like texture and photorealism is less regular than more idealised models.



Fig. 6. Image Generated by StyleGAN

Fig. 6. illustrates the output generated by the StyleGAN framework. By leveraging a mapping network and progressive synthesis, StyleGAN produces high-resolution and photorealistic face images. The generated samples show significant improvement in facial detail, realism, and diversity compared to Conditional DCGAN, while maintaining consistency with the conditioning prompts.

VI. RESULTS AND DISCUSSIONS

The StyleGAN and cDCGAN are evaluated experimentally, and it can be seen that these two methods have a great number of differences in their performance regarding the image generation task. Frechet Inception Distance (FID) and Inception Score (IS) are quantitative values that are used to objectively assess the quality of the model [5], [7]. In the same way, the Inception Score of StyleGAN was 17.83, more than cDCGAN, which proves that StyleGAN not only generates more realistic images but also preserves a greater variety of generated samples [7], [9]. Additionally, it indicates that StyleGAN is always diverse as well as semantically consistent between generated face samples [7], [9].

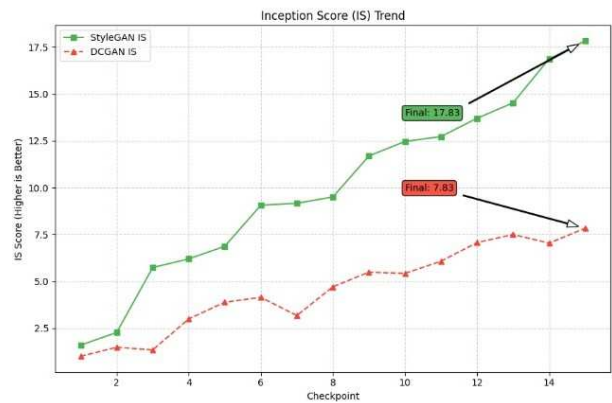


Fig. 7. Inception Score

The IS of cDCGAN in the fig. 7. is extremely high (17.83), which means that the results of cDCGAN are highly diverse and varied [5], [7]. However, this diversity is not always realistic since it has been confirmed by its low FID score [7]. StyleGAN on the other hand has an IS of 7.83 that is not very high but effective in text-to-face generation tasks yet, its FID results are better [7], [9]. This reduced IS and high FID signify that StyleGAN produces more lifelike and consistent images even though it does not produce such a variety of images in comparison to cDCGAN [7], [9].

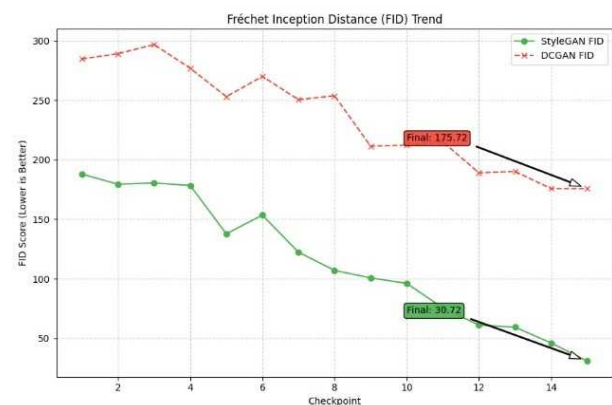


Fig. 8. Fr chet Inception Distance

According to the trend of the FID in the fig. 8., one can mention that StyleGAN can achieve a stable and significant decrease in FID on all training checkpoints and finish the final

value with 30.72 [7], [9]. This is a low FID score, which implies that the images generated by StyleGAN are highly realistic, and that distribution resembles that of the real dataset [7], [9]. Quite to the contrary, cDCGAN has a slower and less steady increase because it reaches a much higher FID of 175.72 [5], [7]. It is an indication of reduced distribution alignment and quality of the image [5], [7]. The gap between the two curves is growing over time which points to the superior learning and generating capacity of StyleGAN toward this dataset [7], [9].

VII. CONCLUSION

We have compared the modern architecture of Conditional DCGAN and StyleGAN in this paper to the task of synthetic human face generation under prompt-based conditions [5], [7], [9]. Experimental results based on both qualitative observations and quantitative scores like Inception Score (IS) and Frechet Inception Distance (FID) showed that StyleGAN was more and more effective than Conditional DCGAN on photorealism, diversity, and prompt alignment [7], [9]. Although Conditional DCGAN offered a low-weight baseline that can learn coarse semantic attributes, it tended to generate artifacts and was not detailed [5], [7]. StyleGAN, in turn, generated high-resolution and visually consistent face images with high levels of consistency with the conditioning cues [7], [9].

The results indicate that StyleGAN can be used more successfully in the generation of large-scale artificial data that may be used in research in face recognition, image translation, and human-AI interaction [7], [9]. Nevertheless, there are still issues like high cost of computation and lack of control on certain semantic attributes [7]. The further development of work will include the incorporation of more advanced text encoders, hybrid GAN-diffusion frameworks, and real-world benchmark tests, which will help to make the synthetic face generation system more realistic, controllable, and less biased [9], [14].

It is anticipated that in the future, hybrid GAN-diffusion models can be used to improve the controllability and visual fidelity of text-to-face generation, which will allow producing more subtle and semantically faithful results [9]. The other crucial direction is that identity stability in personalized generation should be maintained whereby synthetic faces show consistent features across prompts and sessions [12], [13], [15]. Beyond this, the generative model generality of models like StyleGAN has already spread beyond human faces into highly specific tasks, such as medical imaging (e.g., endoscopic image enhancement) and material texture synthesis (e.g., high-quality wood textures), demonstrating that these architectures can be applied to both creative and entertainment tasks as well as important tasks, like healthcare and industry [9], [10], [11].

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